Cambridge (CIE) A Level Maths: Probability & Statistics 1

Your notes

Further Probability

Contents

- Set Notation & Conditional Probability
- Venn Diagrams with Conditional Probability
- Tree Diagrams with Conditional Probability
- Probability Formulae

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Your notes

Set Notation

What is set notation?

- **Set notation** is a formal way of writing groups of numbers (or other mathematical entities such as shapes) that share a common feature – each number in a set is called an **element** of the set
  - You should have come across common sets of numbers such as the **natural numbers**, denoted by $\mathbb{N}$, or the set of **real numbers**, denoted by $\mathbb{R}$
- In **probability**, set notation allows us to talk about the **sample space** and **events** within in it
  - $\mathscr{E}$, S, U and ξ are common symbols used for the **Universal set**
    In probability this is the entire **sample space**
  - **Events** are denoted by capital letters, *A, B, C* etc
  - *A'* is called the **complement** of and means "*not A*"
    (Strictly pronounced " *A* prime" but often called "*A* dash")
    - Recall the important and easily missed result $\mathbf{P}(A') = 1 - \mathbf{P}(A)$
  - **AND** is denoted by ∩ (**intersection**)
    OR is denoted by ∪ (**union**) (remember $A \cup B$ includes both)
- The other set you may come across in probability is the **empty set**
  The **empty set** has no **elements** and is denoted by ∅

The **intersection** of **mutually exclusive** events is the empty set,

- Set notation allows us to write probability results formally
  - For **independent** events: $\mathrm{P}(A \cap B) = \mathrm{P}(A) \times \mathrm{P}(B)$
  - For **mutually exclusive** events: $\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B)$

How do I solve problems given in set notation?

- Recognise the notation and symbols used and then interpret them in terms of **AND** ($\cap$), **OR** ($\cup$) and/or **NOT** (') statements
- Venn diagrams lend themselves particularly well to deducing which sets or parts of sets are involved- draw mini-Venn diagrams and shade them

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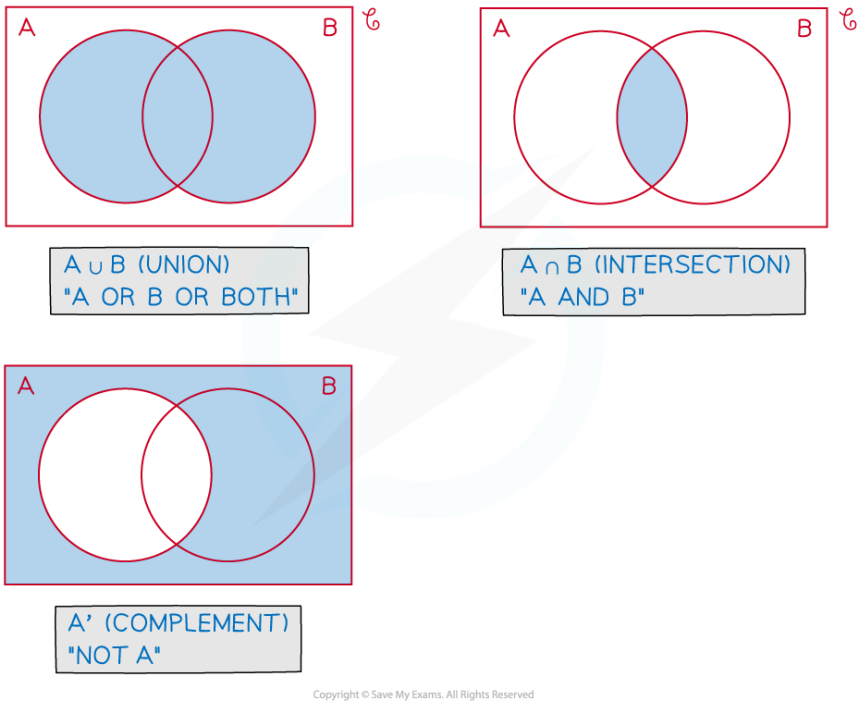
Your notes



- Practice shading various parts of Venn diagrams and then writing what you have shaded in set notation
- With combinations of **union**, **intersection** and **complement** there may be more than one way to write the set required
  - e.g. $(A \cup B)' = A' \cap B'$
    $(A \cap B)' = A' \cup B'$
    Not convinced? Sketch a Venn diagram and shade it in!
  - In such questions it can be the **unshaded** part that represents the solution

Worked Example

The members of a local tennis club can decide whether to play in a singles competition, a doubles competition, both or neither.

Once all members have made their choice the chairman of the club selects, at random, one member to interview about their decision.

S is the event a member selected the singles competition.

D is the event a member selected the doubles competition.

Given that $\mathrm{P}(S) = 2\mathrm{P}(D)$, $P(S \cup D) = 0.9$ and $P(S \cap D) = 0.3$, find

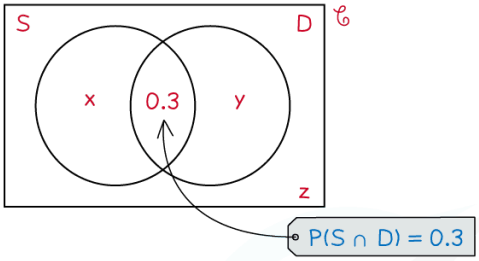
(i) $P(S')$

(ii) $P(S' \cap D)$

(iii) $P(S \cup D')$

(iv) $P((S \cup D)')$

Sketch a Venn diagram and use it to help work out the missing probabilities



$P(S) = 2P(D)$ $\quad \therefore x + 0.3 = 2(0.3 + y)$

$x - 2y = 0.3$

$P(S \cup D) = 0.9$ $\quad \therefore x + 0.3 + y = 0.9$

$x + y = 0.6$

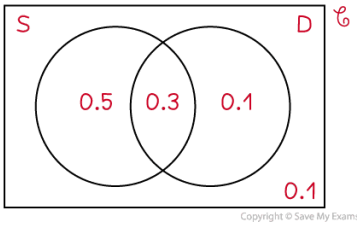
Solve simultaneously, $\quad 3y = 0.3, \quad y = 0.1$

$x = 0.5$

As all probabilities total 1, $0.5 + 0.3 + 0.1 + z = 1$

$z = 0.1$

Final Venn diagram



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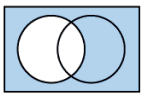
Your notes

Now use this and, if they help, shaded mini-Venn diagrams to answer the questions

(i)



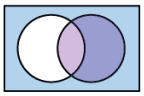
"not S"

$0.1 + 0.1 = 0.2$

$P(S') = 1 - P(S) = 1 - (0.5 + 0.3)$

$P(S') = 0.2$

(ii) $P(S' \cap D)$



Blue shading is "not S
Purple shading is D
$\cap$ is AND, so part shaded twice is the answer

$P(S' \cap D) = 0.1$

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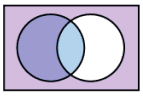
(iii) $P(S \cup D')$

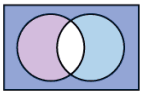


Blue shading is S
Purple shading is "not D"
$\cup$ is OR, so all shaded parts are needed

$P(S \cup D') = 0.5 + 0.3 + 0.1 = 0.9$
(You may also spot from the mini-Venn diagram this is $1 - 0.1$)

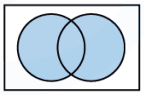
(iv) $P((S \cup D)')$

Method 1
$P((S \cup D)') = P(S' \cap D')$



Answer is the region shaded twice

Method 2
Start by shading $S \cup D$



$(S \cup D)'$ is "not $S \cup D$" so the answer is the unshaded region

$P((S \cup D)') = 0.1$

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Examiner Tips and Tricks

- Venn diagrams are not expected but they are extremely useful
  - Do not try to do everything on one diagram though - use mini-Venn diagrams with shading (no values) for each part of a question
- Do double check whether you are dealing with **union** ($\cup$) or **intersection** ($\cap$) (or both) – when these symbols are used several times near each other in a question, it is easy to get them muddled up or misread them

Conditional Probability

Your notes

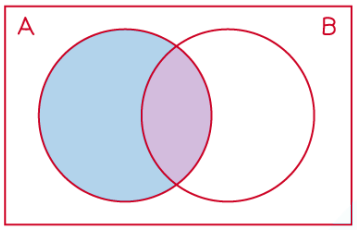
What is conditional probability?

Your notes

- You have already been using **conditional probability in Tree Diagrams**
  - **Probabilities change** depending on the outcome of a prior event
- Consider the following example

e.g. Bag with 6 white and 3 red buttons. One is drawn at random and not replaced. A second button is drawn. The probability that the second button is white **given that** the first button is white is $\frac{5}{8}$.

- The key phrase here is "**given that**" – it essentially means something has already happened.
  - In set notation, "**given that**" is indicated by a vertical line (**|**) so the above example would be written $P(2^{nd} \text{ is white} \mid 1^{st} \text{ is white}) = \frac{5}{8}$
  - There are other phrases that imply or mean the same things as "given that"
- Tree diagrams are great for events that follow on from one another
  - Otherwise **Venn diagrams** are extremely useful
    Beware! The denominator of fractional probabilities will no longer be the total of all the frequencies or probabilities shown
  - "**given that**" questions usually reduce the **sample space** as an event (a subset of the outcomes of the first experiment) has already occurred

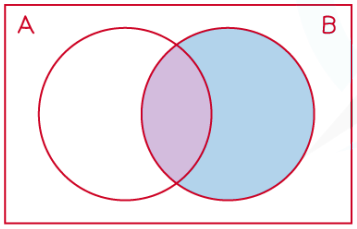


BLUE SHADING IS EVENT A

PURPLE SHADING IS EVENT B, GIVEN THAT EVENT A HAS ALREADY OCCURRED

$\therefore P(B \mid A) = \frac{P(A \cap B)}{P(A)}$

B | A
"B, GIVEN A"



BLUE SHADING IS EVENT B

PURPLE SHADING IS EVENT A, GIVEN THAT EVENT B HAS ALREADY OCCURRED

$\therefore P(A \mid B) = \frac{P(A \cap B)}{P(B)}$

A | B
"A, GIVEN B"

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- The diagrams above also show two more **conditional probability** results
  - $\mathrm{P}(A \cap B) = \mathrm{P}(A) \times \mathrm{P}(B|A)$

  $\mathrm{P}(A \cap B) = \mathrm{P}(B) \times \mathrm{P}(A|B)$

(These are essentially the same as letters are interchangeable)

- For **independent events** we know $\mathrm{P}(A \cap B) = \mathrm{P}(A) \times \mathrm{P}(B)$ so

$$\mathrm{P}(B|A) = \frac{\cancel{\mathrm{P}(A)} \times \mathrm{P}(B)}{\cancel{\mathrm{P}(A)}} = \mathrm{P}(B)$$

and similarly

$$\mathrm{P}(A|B) = \mathrm{P}(A)$$

- The independent result should make sense logically – if events *A* and *B* are independent then the fact that event *B* has already occurred has no effect on the probability of event *A* happening

Your notes

Worked Example

The probabilities of two events, A and B are described as $\mathrm{P}(A) = 0.4$ and $\mathrm{P}(B) = 0.5$.
It is also known that $P(A \cap B) = 0.2$.

(a)

Find

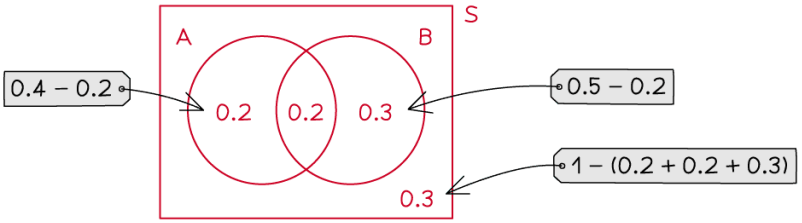
(i) $P(A|B)$

(ii) $P(B|A')$

(iii) $P(A \cap B)(A \cup B)$

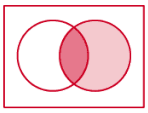
(b) Show, in two different ways, that the events A and B are independent.

a) A Venn diagram is useful here



Use mini–Venn diagrams with shading to help

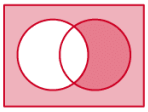
(i)



$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.2}{0.5}$$

$$P(A \mid B) = 0.4$$

(ii)

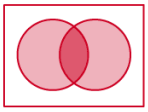


$$P(B \mid A') = \frac{P(B \cap A')}{P(A')} = \frac{0.3}{1 - 0.4}$$

$P(A') = 1 - P(A)$

$$P(B \mid A') = 0.5$$

(iii)



$$P((A \cap B) \mid (A \cup B)) = \frac{P((A \cap B) \cap (A \cup B))}{P(A \cup B)}$$

$$= \frac{0.2}{0.2 + 0.2 + 0.3}$$

or $1 - 0.3$

$$P((A \cap B) \mid (A \cup B)) = \frac{2}{7}$$

b) For independent events "$P(A \mid B) = P(A)$" and "$P(A \cap B) = P(A) \times P(B)$"

$P(A) = 0.4 = P(A \mid B)$ ($P(A \mid B)$ from part (a))

$P(A) \times P(B) = 0.4 \times 0.5 = 0.2 = P(A \cap B)$

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Your notes

Examiner Tips and Tricks

- There are now several symbols used from set notation in probability – make sure you are familiar with them
  - **union** ($\cup$)
  - **intersection** ($\cap$)
  - **not** (')
  - **given that** (|)
- Use Venn diagrams to help deduce missing probabilities in questions – you may find it easier to work these out first before answering questions directly

Two-Way Tables

What are two-way tables?

- In **probability**, **two-way tables** list the frequencies for the outcomes of two events – one event along the top (columns), one event down the side (rows)

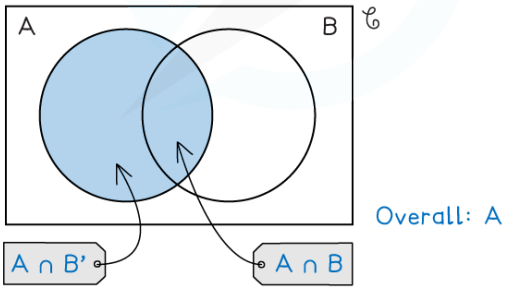
- The frequencies, along with a "Total" row and "Total" column instantly show the values involved in finding probabilities

Your notes

For two events, A and B,

	A	A'	Total
B	$A \cap B$	$A' \cap B$	B
B'	$A \cap B'$	$A' \cap B'$	B'
Total	A	A'	$\mathscr{E}$

Try sketching a Venn diagram for each 'total' cell.
Here's the one for A:



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How do I solve problems given involving two-way tables?

- Questions will usually be wordy – and may not even mention two-way tables
  - Questions will need to be interpreted in terms of **AND** ($\cap$, **intersection**), **OR** ($\cup$, **union**), **NOT** (') and **GIVEN THAT** (|)
- Complete as much of the table as possible from the information given in the question
  - If any empty cells remain, see if they can be calculated by looking for a row or column with just one missing value
- Each cell in the table is **similar** to a region in a Venn diagram
  - With event *A* outcomes on columns and event *B* outcomes on rows
    - $P \cap Q$ (**intersection, AND**) will be the cell where outcome meets outcome Q
    - $P \cup Q$ (**union**, **OR**) will be all the cells for outcomes *P* and Q including the cell for both
  - Beware! As **union** includes the cell for **both** outcomes, avoid counting this cell **twice** when calculating **frequencies** or **probabilities**

(see Worked Example Q(b)(ii))

- You may need to use the results
  - $\mathrm{P}(A \cap B) = \mathrm{P}(A) \times \mathrm{P}(B \mid A)$

- $P(A|B) = P(A)$ (for independent events)

Your notes

Worked Example

The incomplete two-way table below shows the type of main meal provided by 80 owners to their cat(s) or dog(s).

	Dry Food	Wet Food	Raw Food	**Total**
Dog	11		8	
Cat		19		33
Total	21			

(a) Complete the two-way table

(b) One of the 80 owners is selected at random.
Find the probability

(i) the selected owner has a cat and feeds it raw food for its main meal.

(ii) the selected owner has a dog or feeds it wet food for its main meal.

(iii) the owner feeds raw food to its pet, given it is a dog.

(iv) the owner has a cat, given that they feed it dry food.

a) The key valve here is the grand total of 80 – given in the wording of the question

	Dry food	Wet food	Raw food	Total
Dog	11	28	8	47
Cat	10	19	4	33
Total	21	47	12	80

b) (i) $P(\text{cat} \cap \text{raw}) = \frac{4}{80}$

(ii) $P(\text{dog} \cup \text{wet}) = \frac{47}{80} + \frac{47}{80} - \frac{28}{80} = \frac{33}{40}$

P(dog)

P(wet)

Beware! P(dog ∩ wet) would've been counted twice

(iii) $P(\text{raw} \mid \text{dog}) = \frac{8}{47}$

'Dog' 'raw food'

It's given "it's a dog"

(iv) $P(\text{cat} \mid \text{dry}) = \frac{10}{21}$

'Cat' 'dry food'

It's given "dry food"

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Your notes

Examiner Tips and Tricks

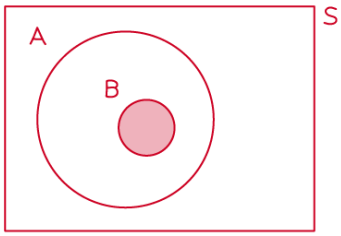
- Ensure any table – given or drawn - has a "Total" row and a "Total" column
- Do not confuse a two-way table with a **sample space** diagram – a two-way table does not necessarily display **all outcomes** from an **experiment**, just those (**events**) we are interested in

Your notes

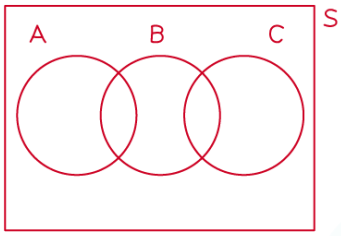
Further Venn Diagrams

What do you mean by further Venn diagrams?

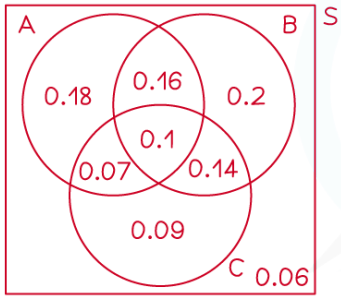
- The Venn diagrams used here are no more complicated than those in the first **Venn Diagrams** revision note, however
  - questions may use **set notation** as well, or alongside contextual questions
  - there may be more detailed use of **conditional probability**
  - **mutually exclusive**, three events and other unusual Venn diagram setups may be involved



THE BUBBLE FOR EVENT B LIES ENTIRELY IN THE BUBBLE FOR EVENT A
IF EVENT B OCCURS SO DOES A
(BUT NOT NECESSARILY VICE VERSA)
$P(A \mid B) = 1$

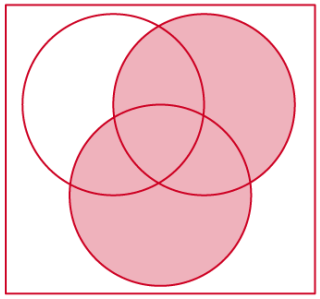


THE BUBBLE FOR EVENT A AND C DO NOT OVERLAP
A AND C ARE MUTUALLY EXCLUSIVE



THREE EVENTS, ALL OVERLAPPING
$P(A) = 0.18 + 0.16 + 0.1 + 0.07 = 0.51$
$P(A \cap B \cap C) = 0.1$
$P(A \cap B) = 0.16 + 0.1 = 0.26$
$P(B \cup C) = 1 - 0.6 - 0.18 = 0.76$

MINI-VENN DIAGRAMS WITH SHADING ONLY ARE VERY USEFUL IN MORE COMPLICATED SCENARIOS



FOR ABOVE, $P(B \cup C)$

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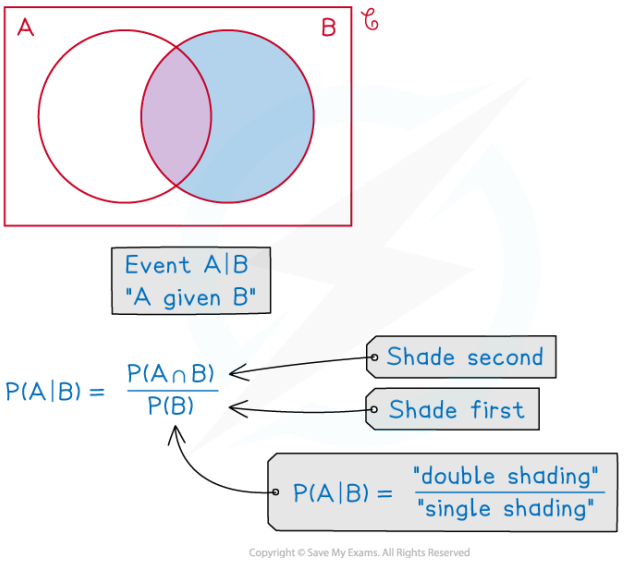
How do I solve conditional probability problems using Venn diagrams?

Your notes

- Interpreting questions in terms of **AND** ($\cup$), **OR** ($\cap$), **complement** (') and **"given that"** (|)
- Use mini-Venn diagrams to sketch and shade the regions you are dealing with – use different colours if available or different styles of shading if not
  - Shading can help you see the answer

since $\mathrm{P}(A|B) = \dfrac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$ shade B first, then shade $A \cap B$

the answer will then be $\dfrac{\text{" double shading "}}{\text{" single shading "}}$



Worked Example

Three events, A, B and C are such that

events A and C are mutually exclusive

$$\mathrm{P}(A \cap B) = 0.2$$
$$\mathrm{P}(B \cap C) = 0.3$$
$$\mathrm{P}((A \cup B \cup C)') = 0.1$$
$$\mathrm{P}(B) = 0.7$$
$$\mathrm{P}(A') = 0.75.$$

Find

(i) $P(A \mid B)$

(ii) $P(A' \mid C')$

(iii) $P(C \mid (A \cup B)')$

Sketch a Venn diagram from the information given

A, C mutually exclusive – bubbles do not overlap

P(A∩B) = 0.2 – "A AND B"

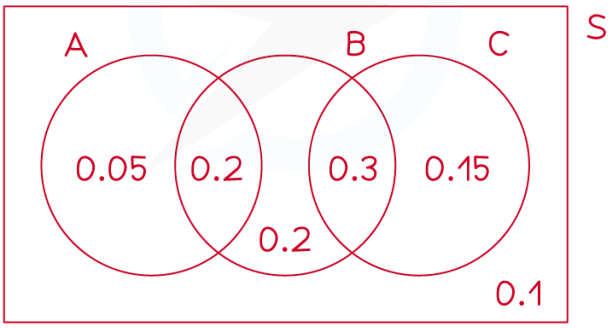
P(B∩C) = 0.3 – "B AND C"

P((A∪B∪C)') = 0.1 – "NOT (A OR B OR C)"

P(B) = 0.7 – "ONLY B" = 0.2

P(A)' = 0.75 – "ONLY A" = 0.05

and SO "ONLY C" = 0.15



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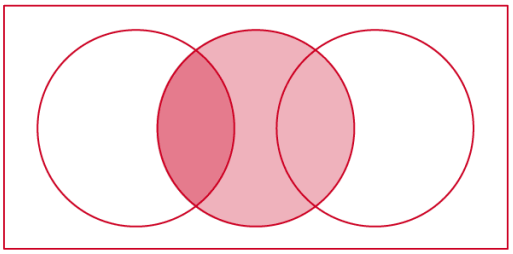
Your notes

(i) $P(A \mid B)$

Mini – Venn:



$P(A \mid B) = \frac{2}{7}$

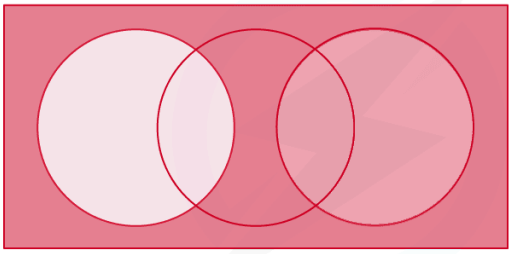
Formula:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$= \frac{0.2}{0.7}$$

(ii) $P(A' \mid C')$

Mini – Venn:



$P(A' \mid C') = \frac{6}{11}$

Double shading

Formula:

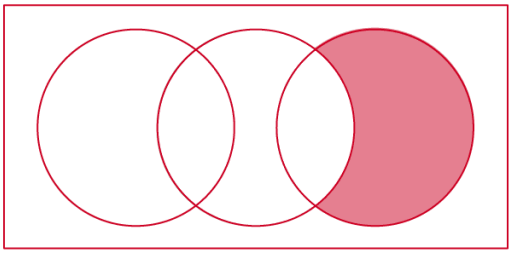
$$P(A' \mid C') = \frac{P(A' \cap C')}{P(C')}$$

$$= \frac{0.2 + 0.1}{1 - 0.45}$$

Light–pink shading

(iii) $P(C \mid (A \cup B)')$

Mini – Venn:



$P(C \mid (A \cup B)') = \frac{3}{5}$

Formula:

$P(C') \mid (A \cup B)')$

$$= \frac{P(C \cap (A \cup B)')}{P((A \cup B)')}$$

$$= \frac{0.15}{0.15 + 0.1}$$

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Your notes

Examiner Tips and Tricks

- Although very versatile, Venn diagrams may not always be the best option
  - use a tree diagram if one event follows another

e.g. two beads drawn from a bag without replacement

- use a two-way table for showing how many items/people are in associated categories
  e.g. A school year group split by gender selecting which sport they wish to play

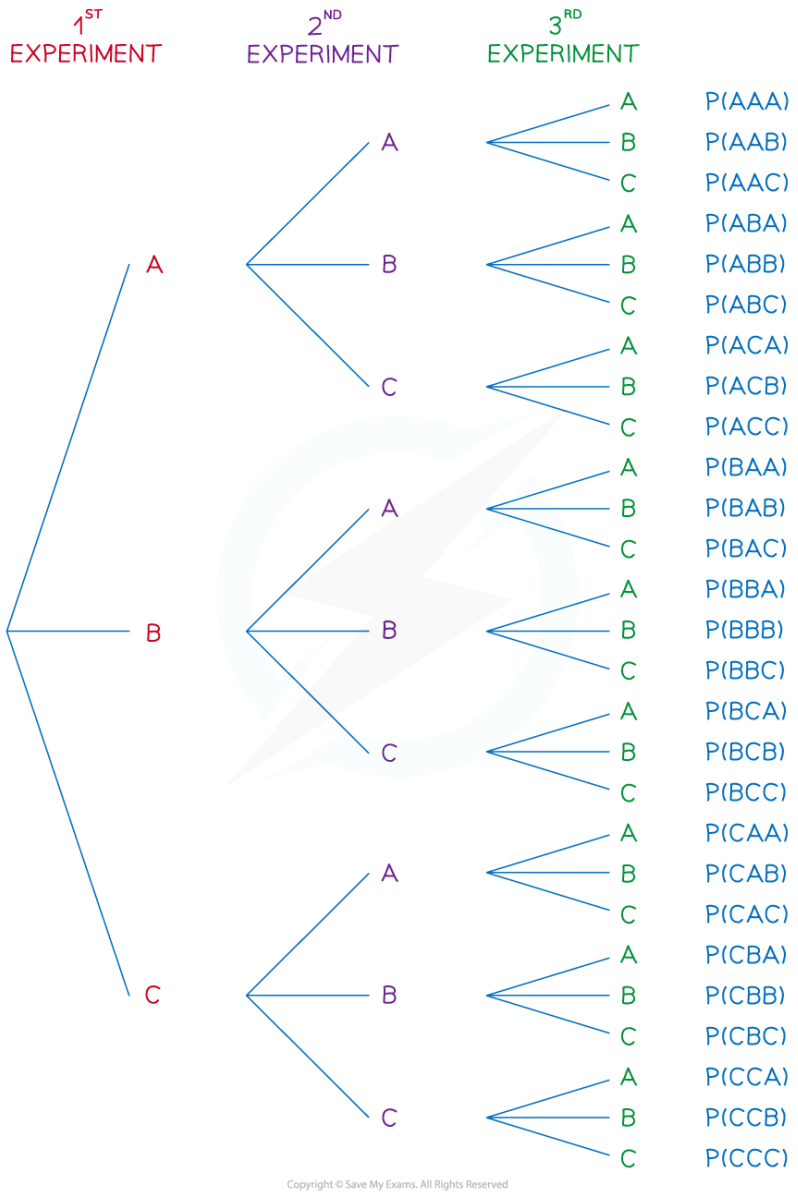
Your notes

Your notes

Further Tree Diagrams

What do you mean by further tree diagrams?

- The tree diagrams used here are no more complicated than those in the first **Tree Diagrams** revision note, however
  - questions may use **set notation** as well, or alongside contextual questions $\cup$ (**union**), $\cap$ (**intersection**) , ' (**complement**), | ("**given that**")
  - more detailed use of **conditional probability**
  - three **events** for each **experiment** and three experiments could be used



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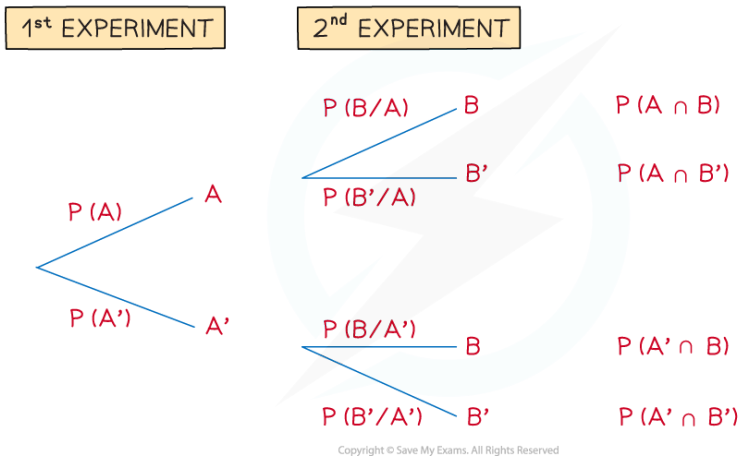
How do I solve conditional probability problems using tree diagrams?

Your notes

- Interpreting questions in terms of **AND** ($\cap$), **OR** ($\cup$), **complement** (') and **"given that"** (|)
- **Condition probability** may now be involved too
- This makes it harder to know where to start and how to complete the probabilities on a tree diagram
  - e.g. If given, possibly in words, $P(B|A)$ then **event *A*** has already occurred so start by looking for the branch **event *A*** in the **1st experiment,** and then P(B | A) would be the branch for **event *B*** in the **2nd experiment**

Similarly, $P(B|A')$ would require starting with **event "*not A*"** in the **1st experiment** and **event *B*** in the **2nd experiment**



- The diagram above gives rise to some probability formulae you will see in Probability Formulae
- $\mathbf{P}(B|A)$ ("**given that**") is the probability on the branch of the 2nd experiment
- However, the "**given that**" statement $\mathbf{P}(A|B)$ is more complicated and a matter of working backwards
  - from Conditional Probability, $\mathrm{P}(A|B) = \dfrac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$
  - from the diagram above, $P(B) = P(A \cap B) + P(A' \cap B)$
  - leading to $\mathbf{P}(A|B) = \dfrac{\mathbf{P}(\mathrm{A} \cap \mathrm{B})}{\mathbf{P}(\mathrm{A} \cap \mathrm{B}) + \mathbf{P}(\mathrm{A}' \cap \mathrm{B})}$
  - This is quite a complicated looking formula to try to remember so use the logical steps instead – and a clearly labelled tree diagram!

Your notes

Worked Example

The event F has a 75% probability of occurring.

The event W follows event F, and if event F has occurred, event W has an 80% chance of occurring.

It is also known that $P(F' \cap W) = 0.15$.

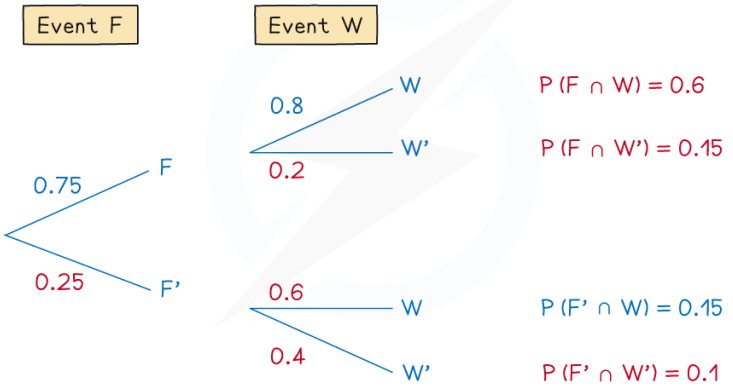
Find

(i) $\mathrm{P}(W|F')$

(ii) $\mathrm{P}(F|W')$

(iii) the probability that event F didn't occur, given that event W didn't occur.

Drawing a tree diagram will help determine values in probabilities, you can then focus on interpreting the operations



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i) $P(W|F')$ F' on "1st experiment', W on "2nd"

$\therefore P(W|F') = 0.6$

ii) $P(F|W')$ Use "$P(A|B) = \frac{P(A \cap B)}{P(B)}$" and logic...

$P(F \cap W') = 0.15$

$P(W') = P(F \cap W') + P(F' \cap W')$

$= 0.15 + 0.4 = 0.55$

$\therefore P(F|W') = \frac{0.15}{0.55}$

$\therefore P(F|W') = \frac{3}{11}$

iii) This is the same as $P(F'|W')$ and so a similar formula and logic would be recommended

But to show how the formula in the revision note would apply, we've used that below

$$P(F'|W') = \frac{P(F' \cap W')}{P(F \cap W') + P(F' \cap W')}$$

$$= \frac{0.1}{0.15 + 0.1}$$

$$= \frac{0.1}{0.25}$$

$\therefore P(F'|W') = 0.4$

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Your notes

Examiner Tips and Tricks

- Be wary of assuming that "given that" statements will always be referring to something on the second set of branches (2nd experiment), they can work the other way!

Your notes

Probability Formulae

What is meant by probability formulae?

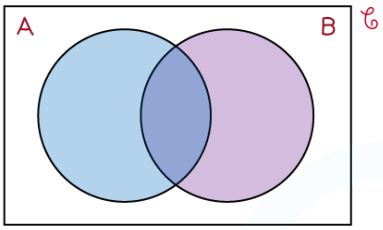
- If you have seen any of the other Revision Notes on probability, almost every one contains a formula
  - These however are often disguised in Venn diagrams, tree diagrams etc
  - Sometimes a diagram can be easier to use than a formula
- This Revision Note rounds up all these formulae and introduces one or two new ones
- You are not expected to use these formulae and none are given in the formulae booklet but they can help solve problems

What probability formulae are there?

- Some formulae only apply under certain conditions
- The formulae for probability you should've encountered so far
  - for **independent** events, $\mathrm{P}(A \cap B) = \mathrm{P}(A) \times \mathrm{P}(B)$ **("AND", intersection)**
  - for **mutually exclusive** events, $\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B)$ **("OR", union)**
  - $\mathrm{P}(A') = 1 - \mathrm{P}(A)$ **("NOT", complement/prime/dash)**
  - $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$ or $\mathrm{P}(A \cap B) = \mathrm{P}(B) \times \mathrm{P}(A \mid B)$
    ("GIVEN THAT")
    This second version in particular is referred to as the **MULTIPLICATION FORMULA** (see diagram below)
  - For **independent** events, $\mathrm{P}(A \mid B) = \mathrm{P}(A)$

Your notes

Consider the shaded parts of the Venn diagram



P(A) is shaded blue
P(B) is shaded purple
P(A∩B) is shaded twice
P(A∪B) is shaded at least once

From this we can deduce
P(A ∪ B) = P(A) + P(B) − P(A ∩ B)

Subtract the region that gets counted twice

This is called the ADDITION FORMULA
It can easily be rearranged to

P(A ∩ B) = P(A) + P(B) − P(A ∪ B)

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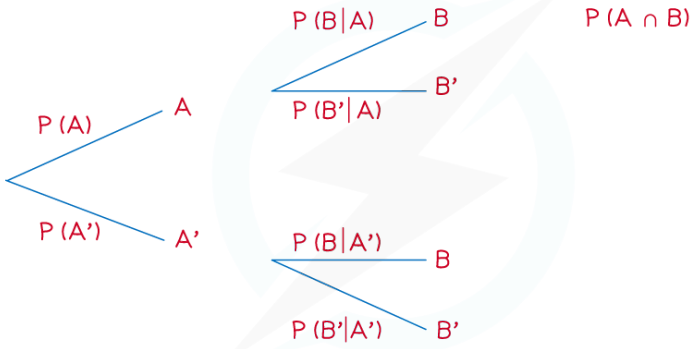
There is also a MULTIPLICATION FORMULAE
Consider the top branches (or any pair really) of this tree diagram



So we can deduce

P(A ∩ B) = P(A) × P(B|A)

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- The **addition formula** can be rearranged in two very similar ways so be careful
  - $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ and

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

- The special case is when A and B are **mutually exclusive**; in such cases then

$P(A \cap B) = 0$ and so $P(A \cup B) = P(A) + P(B)$

How do I solve problems using probability formulae?

Your notes

- This is a combination of
  - Recognising the **set notation** used
  - Taking note of any **independent** or **mutually exclusive** events
  - Using an appropriate diagram (Venn, mini-Venn, tree, two-way table)
  - Converting worded questions into **AND**, **OR**, **GIVEN THAT** etc statements
  - Knowing when a formula could be used
- In the majority of cases a diagram can be used – so if you do not like using the formulae or find them confusing
  - For events that happen in succession, use a tree diagram; Venn diagrams suit most other purposes but two-way tables can be easier to follow

Worked Example

In the fictional World Stare Out Championships players compete by staring at each other with the player blinking first losing. A match cannot be drawn.

During each day of the championships three matches are scheduled to take place but if the first two matches both take more than an hour each, then the third match cannot take place that day.

A statistician notices from past fictional championship records that the probability of the first match taking longer than an hour is 0.15 and that the probability of only two matches taking place on any day is 0.06. They also notice that the probability that at least one of the first two matches takes longer than an hour is 0.32.

Find the probability that

(i) the second match of a day takes longer than an hour to complete

(ii) the first match takes longer than an hour given that the second match takes longer than an hour

Your notes

Let M_i be the event match i takes more than an hour to complete
The question tells us

$P(M_1) = 0.15$ (first match more than an hour)

$P(M_1 \cap M_2) = 0.06$ (only 2 matches – i.e both 1st two matches take more than an hour)

$P(M_1 \cup M_2) = 0.32$ (at least 1 match takes longer than an hour)

(i) $P(M_1 \cap M_2)$ is required. Using "$P(A \cap B) = P(A) + P(B) - P(A \cup B)$"

$P(M_1 \cap M_2) = P(M_1) + P(M_2) - P(M_1 \cup M_2)$

$0.06 = 0.15 + P(M_2) - 0.32$

$\therefore P(M_2) = 0.23$

(ii) $P(M_1 \mid M_2)$ is required. Using "$P(A \cap B) = P(B) \times P(A|B)$"

$P(M_1 \cap M_2) = P(M_2) \times P(M_1 \mid M_2)$

$0.06 = 0.23 \times P(M_1 \mid M_2)$

$\therefore P(M_1 \mid M_2) = \frac{6}{23}$

Note:
The context of this question has one experiment (match 2) following another (match 1) so lends itself to a tree diagram.
If you're confident using tree diagrams, drew one and use it. Do be aware that some questions like part (ii) above are not obvious from a diagram and so the formulae are still important. For completness the final tree diagram would be.

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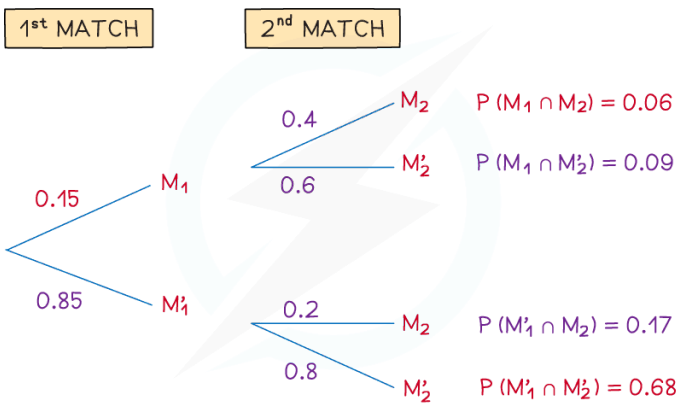

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Examiner Tips and Tricks

- If in any doubt always start with a diagram
  - a **Venn diagram** can be used for most problems
  - a **two-way table** can be easier to read if it's possible to construct one
  - a **tree diagram** is useful if you are looking at an event that follows another

- Remember that all probability formulae are given in the formulae booklet but not necessarily in the most user-friendly way; a quick look could just be enough to jog your memory though!

Your notes

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